

Proposed Solution / System Architecture

Proposed Solution

Date **05 July 2026**

Team ID *********

Project Name **StudyAI – AI Powered Smart Study Companion**

Maximum Marks 3 Marks

Proposed Solution

StudyAI is an AI-powered web application designed to simplify the learning process for students by automatically converting study materials into intelligent learning resources. Instead of manually reading lengthy notes and creating revision content, users can upload their study materials and receive AI-generated summaries, flashcards, quizzes, and personalized study plans.

The system combines Artificial Intelligence with a modern web application architecture to provide an efficient, interactive, and user-friendly learning experience.

Objectives

The primary objectives of StudyAI are:

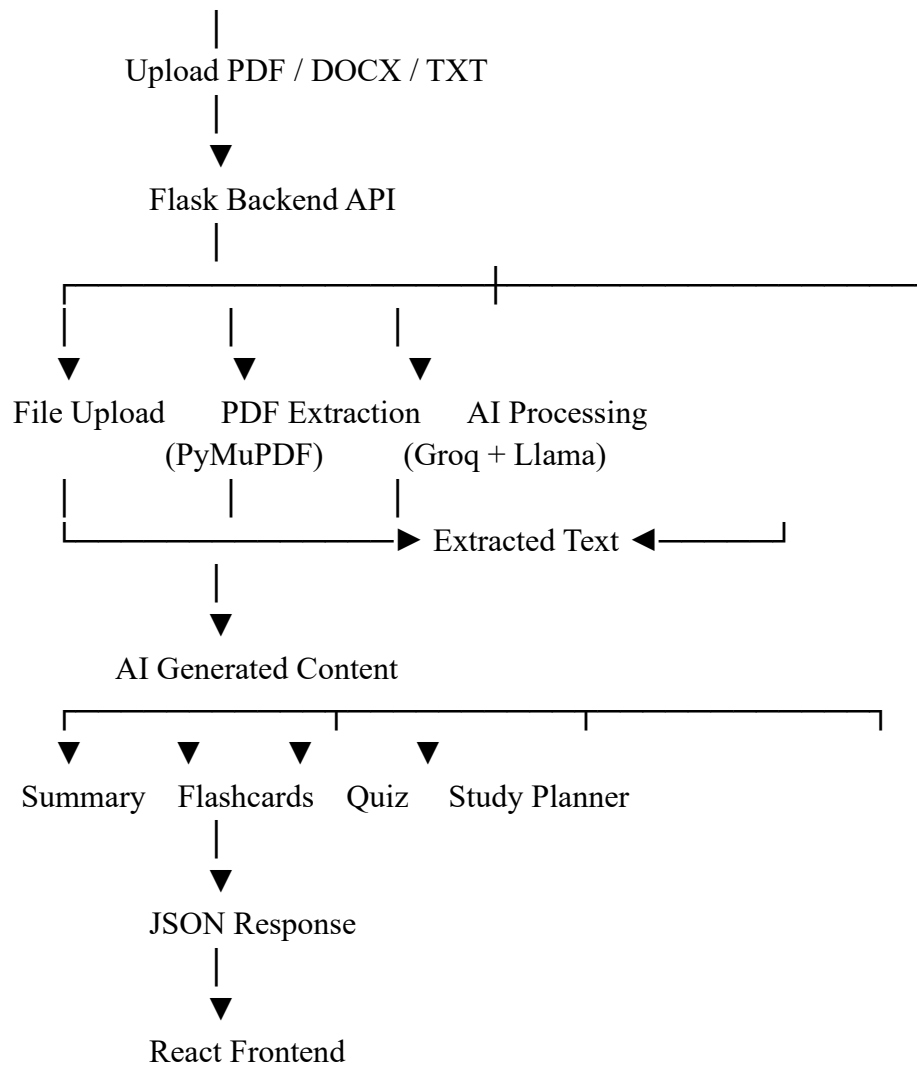
- Reduce the time required for exam preparation.
 - Automate the process of creating study materials.
 - Improve learning through AI-generated content.
 - Provide students with a centralized learning platform.
 - Encourage active recall using quizzes and flashcards.
 - Enhance productivity during revision.
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System Architecture

STUDYAI

React Frontend

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Workflow

Step 1 – Upload Study Material

The user uploads a PDF, DOCX, or TXT file through the React frontend.

Step 2 – File Validation

The Flask backend verifies:

- Supported file type
- Valid filename
- Successful upload

The file is stored with a UUID-based filename to avoid duplication.

Step 3 – Text Extraction

If the uploaded file is a PDF, PyMuPDF extracts the textual content.

Step 4 – AI Processing

The extracted text is sent to the Groq API using the Llama language model.

The AI analyzes the content and generates:

- Concise summaries
 - Flashcards (planned)
 - Quiz questions (planned)
 - Study plan (planned)
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Step 5 – Response Generation

The backend returns the generated content in JSON format.

Step 6 – Display Results

The React frontend displays the AI-generated output to the user.

Module Description

Module 1 – User Interface

Developed using React.js.

Provides pages for:

- Upload
 - Summary
 - Flashcards
 - Quiz
 - Planner
 - Dashboard
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Module 2 – File Upload

Responsible for:

- Uploading study materials.
 - Validating supported file formats.
 - Saving files securely.
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Module 3 – PDF Processing

Uses PyMuPDF to extract readable text from uploaded documents.

Module 4 – AI Engine

Uses Groq API and the Llama model to understand the extracted content and generate intelligent educational resources.

Module 5 – Learning Modules

Displays:

- AI Summary
 - Flashcards
 - Quiz
 - Study Planner
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Advantages of the Proposed Solution

- Faster exam preparation.
 - Reduced manual effort.
 - AI-assisted learning.
 - Improved revision efficiency.
 - Personalized learning experience.
 - Modular and scalable architecture.
 - Easy integration of future AI features.
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Expected Outcome

The proposed StudyAI system enables students to upload study materials and instantly generate meaningful learning resources. By integrating AI into the study workflow, the system minimizes repetitive tasks, improves knowledge retention, and enhances the overall learning experience.

Small improvement

Since **Flashcards, Quiz, and Planner are currently under development**, mention this in your documentation if you're submitting based on the current implementation. It shows honesty and aligns the documentation with the actual state of the project. Claiming those modules are fully implemented when they aren't could create issues during evaluation.